

Cross-Platform Prediction Market Arbitrage: Kalshi vs. Polymarket

Opening

Cross-market arbitrage between Kalshi and Polymarket exists. There are timestamps, the prices, and the fee-adjusted profit figures. But existing and being capturable are two different things.

What began as a manual observation during the Super Bowl turned into an auto detection system logging real opportunities in real time. Every layer I built revealed a new layer of complexity: data delays, fee structures, term mismatches, oracle differences. The conclusion isn't that arbitrage doesn't exist. It's that by the time an independent developer can detect and act on it, the edge has already been extracted.

Phase 1: Manual Observation

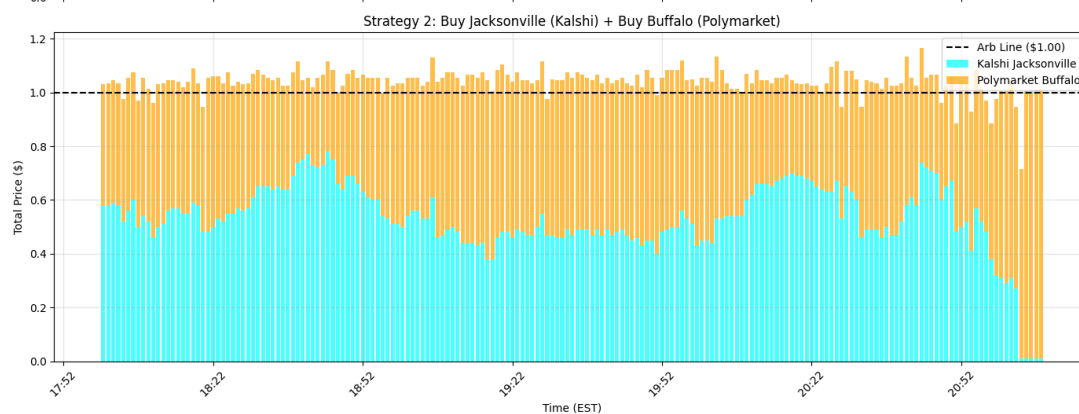
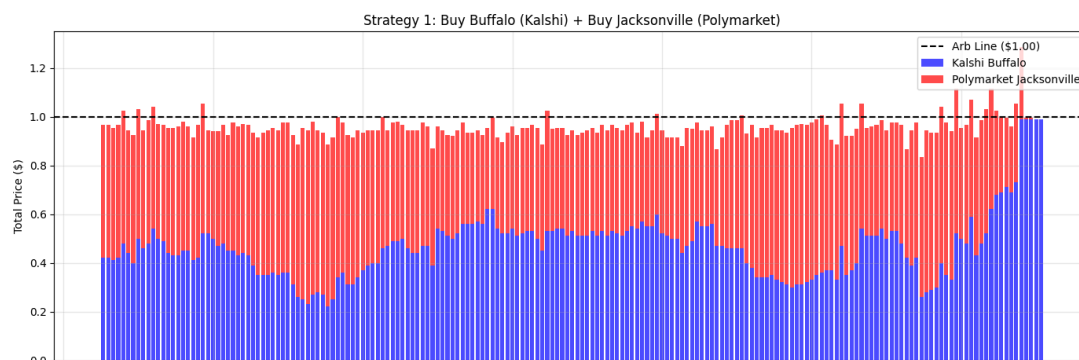
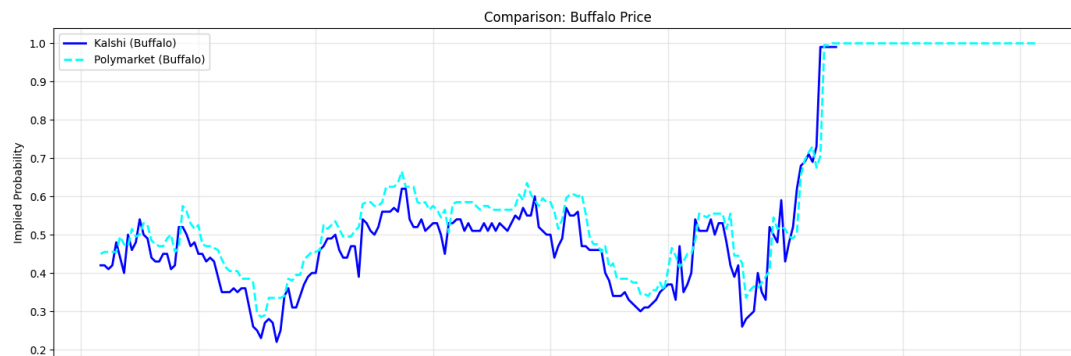
I started with sports markets because the price movements are obvious. A player gets injured, a team scores, and implied probabilities shift visibly. During the Super Bowl and a few basketball games, I noticed a consistent 2–3% gap between Polymarket and Kalshi on the same event.

But manual verification is unreliable. Reloading two browser tabs is not the same as reading two prices simultaneously. Is the gap real, or is it the few seconds between refreshes?

Phase 2: The Lag Discovery

I built data collectors — first historical (minute-by-minute), then live WebSocket feeds. The live data confirmed the gap was real but reframed what kind of gap it was. Polymarket wasn't just priced differently. Kalshi moved first then Polymarket followed.

The explanation is liquidity. During the Jacksonville vs. Buffalo game, Kalshi had \$25M in volume while Polymarket had \$66K on the same event. A thinner book means wider spreads, which makes price changes more expensive to execute, which means they happen more slowly. What looked like mispricing was largely latency.



(An open hypothesis I never tested: does the lag magnitude scale predictably with the volume differential? If so, that ratio might be a useful signal for predicting when and how large the arb window will be.)

Phase 3: Moving to Crypto Markets

Finding markets that had fast turnaround with contracts that had truly matching terms across platforms was harder than expected.

BTC 15-minute up/down markets solved both problems. Both Kalshi and Polymarket list them, both pay \$1 per contract, and they resolve every quarter-hour. But one difference I only noticed a week in... the oracles aren't identical. Kalshi uses CF Benchmarks, a 60-second TWAP. Polymarket uses Chainlink BTC/USD. On smooth moves they track closely. On sharp spikes, the TWAP lags, meaning both legs could resolve the same way. This makes the strategy *soft* arbitrage (high expected value, not guaranteed).

Phase 4: The Detection System

The scanner runs two concurrent WebSocket feeds and a shared arbitrage detector that fires on every price update. It checks two legs: YES @ Kalshi + NO @ Polymarket, and the reverse. For each, it computes costs including taker fees on both sides, and alerts when net profit is positive.

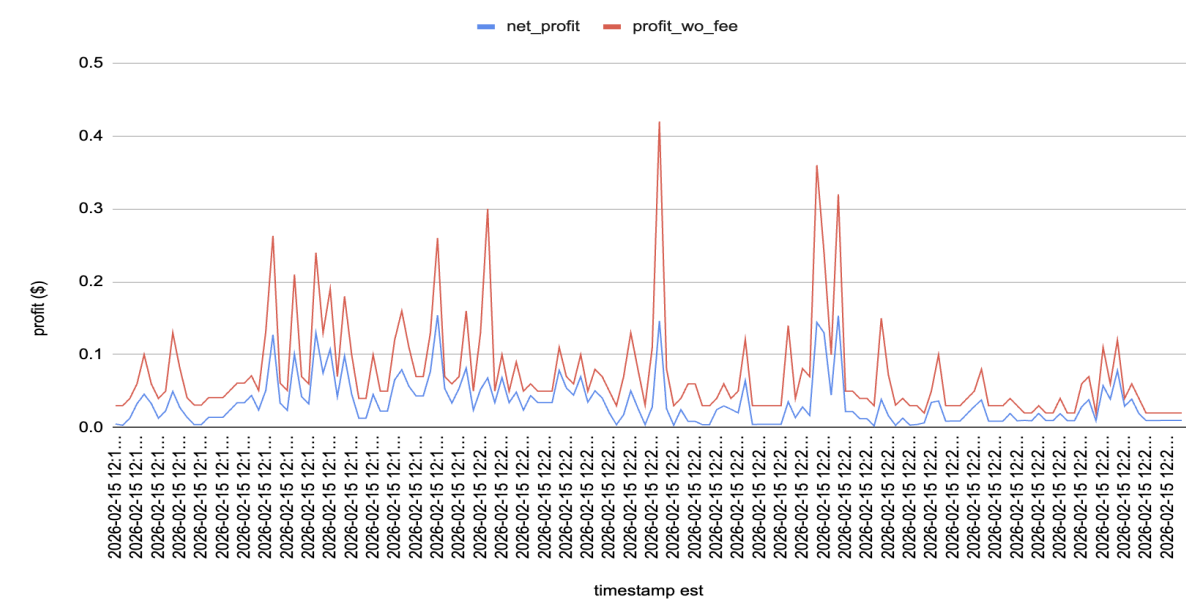
Rather than just checking the best ask, it walks both order ladders simultaneously to estimate how many contracts could actually be filled at a profit, accounting for the fact that a thin book may only support one or two contracts regardless of what the spread implies.

Market rotation is handled automatically. The system parses the current Kalshi ticker expiry, pre-fetches the next Polymarket market config 30 seconds out, and swaps both subscriptions 5 seconds before expiry. The boundary is messy as Polymarket often doesn't publish the next window until close to expiry, and early rotations produce ~\$0.50 false detections from stale prices mixing with a new market's data.

Phase 5: The Fee Reality

Gross profit on detected opportunities was around \$0.03–0.04 per contract. Then fees. Both platforms charge nonlinear taker fees that peak at the 50-cent midpoint:

	Taker	Maker
Kals hi	$\text{fees} = \text{round up}(0.07 \times C \times P \times (1-P))$ P = the price of a contract in dollars (50 cents is 0.5) C = the number of contracts being traded round up = rounds to the next cent	$\text{fees} = \text{round up}(0.0175 \times C \times P \times (1-P))$ P = the price of a contract in dollars (50 cents is 0.5) C = the number of contracts being traded round up = rounds to the next cent
Poly mar ket	$\text{fee} = C \times \text{feeRate} \times p \times (1 - p)$ C = number of shares p = price of shares feeRate = 0.072	$\text{fee} = C \times \text{feeRate} \times p \times (1 - p)$ C = number of shares p = price of shares feeRate = 0



[Visualization of profit before v after fee]

At 50 cents, that's ~3.5¢ combined per contract. Net profit after fees compressed to \$0.01–0.02. This data was from February 2026 when Polymarket's fee rate was capped at ~1.65%. Fees have since increased. Running the same detection under current rates would eliminate most of the remaining signal.

Phase 6: Detection vs. Execution

Every positive number in this analysis is a detected opportunity, not a captured one. If a large fill landed between detection and execution, the estimated max-contracts figure is an overestimate [\[data containing net profit and total available contracts\]](#). A more honest system would paper-trade in real time, measuring the slippage between what the book showed and what would have actually filled.

Beyond my own infrastructure, the competitive landscape had already shifted. By early 2026, institutional desks had entered both platforms and arbitrage windows were compressing to single-digit seconds. A bot open-sourced on Twitter in February 2026 with a note that it was "no longer profitable" marks a useful timestamp for when that transition completed. Detection is achievable at the independent researcher level. Execution against an order book being consumed in under 100ms by institutional infrastructure is a different problem.

Closing

The gap is real. The detection is achievable. The profit, at the infrastructure level available to an independent researcher in 2026, is not.

What this project taught me is something more durable than a trading strategy: how markets close. Every inefficiency that can be systematically detected will attract capital sophisticated enough to eliminate it. The window didn't close because the arbitrage disappeared — it closed because the cost of capturing it dropped to near zero for participants with the right infrastructure, and those participants arrived.
